

Maths for DSA

Date _____
Page _____

Bitwise Operator

(1)	AND	a	b	$a \& b$
		0	0	0
		0	1	0
		1	0	0
		1	1	1

(2)	OR	a	b	$a \mid b$
		0	0	0
		0	1	1
		1	0	1
		1	1	1

(3) XOR (^) (if & only if) Exclusive OR

a	b	$a \wedge b$
0	0	0
1	0	1
0	1	1
1	1	0

$$* a \wedge 1 = \bar{a}$$

$$* a \wedge 0 = a$$

$$* a \wedge a = 0$$

4) Complement (~)

a	$\bar{a}$
0	1
1	0

Number System

(i) Decimal - Base = 10 , 0, 1, 2, ..., 9

(ii) Binary - Base = 2 , 0 & 1

(iii) Octal - Base = 8 , 0-7

(iv) HexaDecimal - Base = 16 , 0-9 & A-F

A = 10 D = 13

B = 11 E = 14

C = 12 F = 15

Some more Operator

5) left shift operator ($\ll$)

$$(10)_{10} = (1010)_2 \quad 10 \ll 1$$

$$\begin{array}{c} \text{nnnn} \\ 1010 \end{array} \ll 1 = \begin{array}{c} \text{nnnn} \\ 10100 \end{array} = 20$$

Note :-

$$* \quad a \ll 1 = 2a$$

$$* \quad a \ll b = a * 2^b$$

6) Right Shift $\gg$

Q. $(0011001)_2 \gg 1 \Rightarrow (001100)_2$

Note :-

* $a \gg 1 = a/2$

* $a \gg b = \frac{a}{2^b}$

Ques 1) Given a number n find its ^{Odd}~~prime~~ or not

$\Rightarrow$ Every no is converted into binary form internally

at 2^0 place $\Rightarrow$

1 is present $\Rightarrow$ odd

0 is present $\Rightarrow$ Even

Eg. $(011101101101)_2$

$\hookrightarrow 1 \Rightarrow$ whole no's odd

So, $n \& 1 == 1$ odd else even.

(2) arr = [2, 3, 4, 1, 2, 1, 3, 6, 4] . Find not duplicates

Sol XOR all number

(3) arr = [-2, 3, 2, 4, -5, 5, -4] . Find no, which have ^{only} +ve

Sol Add all

(4) Find ith bit of a no. ~~1010~~
~~10110110~~ at 5th
~~(00010000)~~ mask $\Rightarrow 1 \ll 4$
~~00010000~~

Ans) num & (1 << (n-1))

~~10110110~~
~~1280000420~~
~~128~~

(4) Find ith bit of a number (5th bit)

M-1
 num = 10110110
 & 00010000 mask
 10000 = ans

So,

if (0) return 0

if (10) return 1.

M-2
 num = 10110110
 num >> (5-1) = 4

00001011

1

00000001 = ans

return ans.

ans = num & (1 << (n-1))

then if (ans != 0)

cout (1)

else

cout (0)

1010
1110
512

Q5) Set i^{th} bit to 1.

4th

Sol^y.

$$1010110$$

$$\text{OR } 0001000 = \text{mask}$$

$$1011110 = \text{ans}$$

Q6) Reset i^{th} bit (to 0)

5th

Sol^y.

$$1010110$$

$$\& 1101111 = \text{mask}$$

$$1000110 \text{ Ans}$$

How to get mask

How to get mask

mask = $\sim (1 < n)$

Q7) Find the position of right most set bit

Sol^y:- $n = 00101101100$

$-n$ = 2's complement of n

$$\begin{array}{r}
 = 010010011 \\
 + \quad \quad \quad 1 \\
 \hline
 010010100
 \end{array}$$

$$\begin{array}{r}
 n \& -n = 101101100 \\
 \quad \quad \quad 010010100 \\
 \hline
 \quad \quad \quad 000000100
 \end{array}$$

AND = 4

position = $\log_2(4) + 1 = 3$

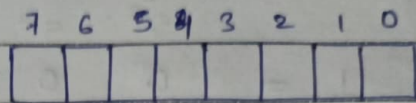
* Range of Numbers

① 1 byte = 8 bit

~~00000000~~

total unique no = 256

(-128) to (127)



↳ it gives positive & negative

* Range formula :- for n bits.

$$-2^{n-1} \text{ to } 2^{n-1} - 1$$

Q8) find the number appearing single time

arr = [2, 2, 3, 2, 7, 7, 8, 7, 8, 8]

• 2 = 10

• 3 = 11

• 7 = 111

• 8 = 1000

Bit 0 = 4 one = $4 \% 3 = 1$

Bit 1 = 4 one = $4 \% 3 = 1$

Bit 2 = 3 one = $3 \% 3 = 0$

Bit 3 = 3 one = $3 \% 3 = 0$

0011 = 3

Q3

Find n^{th} magic number.

		5^3	5^2	5^1	magic no
1	=	0	0	1	5
2	=	0	1	0	25
3	=	0	1	1	30
4	=	1	0	0	125

$$n = 6 \Rightarrow 110$$

$$0^{\text{th}} \text{ Bit} = n \& 1$$

$$0 \times 5^1$$

$$1^{\text{st}} \text{ Bit} = (n \gg 1) \& 1$$

$$1 \times 5^2$$

$$2^{\text{nd}} \text{ Bit} = (n \gg 2) \& 1$$

$$1 \times 5^3$$

$$3^{\text{rd}} \text{ Bit} = (n \gg 3) \& 1$$

$$0 \times 5^4$$

$$\Rightarrow 5^1 + 5^2 = 25 + 125 = 150$$

Q6 Find the no of digit in base

$$(6)_{10} = (110)_2 \Rightarrow 3 \text{ digit}$$

1st method

~~$n \Rightarrow$
 loop {
 $n \neq 1$
 $\text{count}++$;
 $n \gg 1$~~

Formula

$$(\text{int})(\log_b n) + 1$$

Q7 Pascal triangle

```

1
1 1
1 2 1
1 3 3 1
1 4 6 4 1
1 5 10 10 5 1

```

find the sum of n^{th} row

$$\text{sum of each row} = {}^nC_0 + {}^nC_1 + {}^nC_2 + \dots + {}^nC_n = 2^n$$

$$\text{Sum of } n^{\text{th}} \text{ row} = 2^{n-1}$$

$$\text{Ans :- } 1 \leq (n-1) \Rightarrow 1 \times 2^{n-1}$$

Q3. find out if its power to two or not

Method 1

loop { $n \neq 1 = 1$ if count = 1
count + 1 its power of two
 $n > 1$

Method 2

if $n \& (n-1) = 0$ it is power of 2

$n =$
10000000
+ 01111111

0

16
22

$n =$
10101
+ 10100

10100 Not power of two

$n =$
0100
+ 0011

0111

• Find a^b

let $a = 3$

$b = 110$

ans = 1

$b = 3$

$n = 110$

~~if~~ $\left\{ \begin{array}{l} n \neq 1 = 1 \\ \text{ans} *= b \\ b *= b \\ \text{till } n > 0 \end{array} \right. \quad \left\{ \begin{array}{l} n \neq 1 = 1 \\ \text{ans} *= b \\ b *= b \\ n \gg 1 \end{array} \right.$

Q) Given number n ; find no. of set bit in n number

Eg:- $n = 9$

$n = 1001$ ans 2

Method 1 $\Rightarrow$ normal counter. method

Method 2 =

~~while~~

~~and~~ 1

~~while~~

while ($n > 0$) {

$n = (n \& n - 1)$

count++;

}